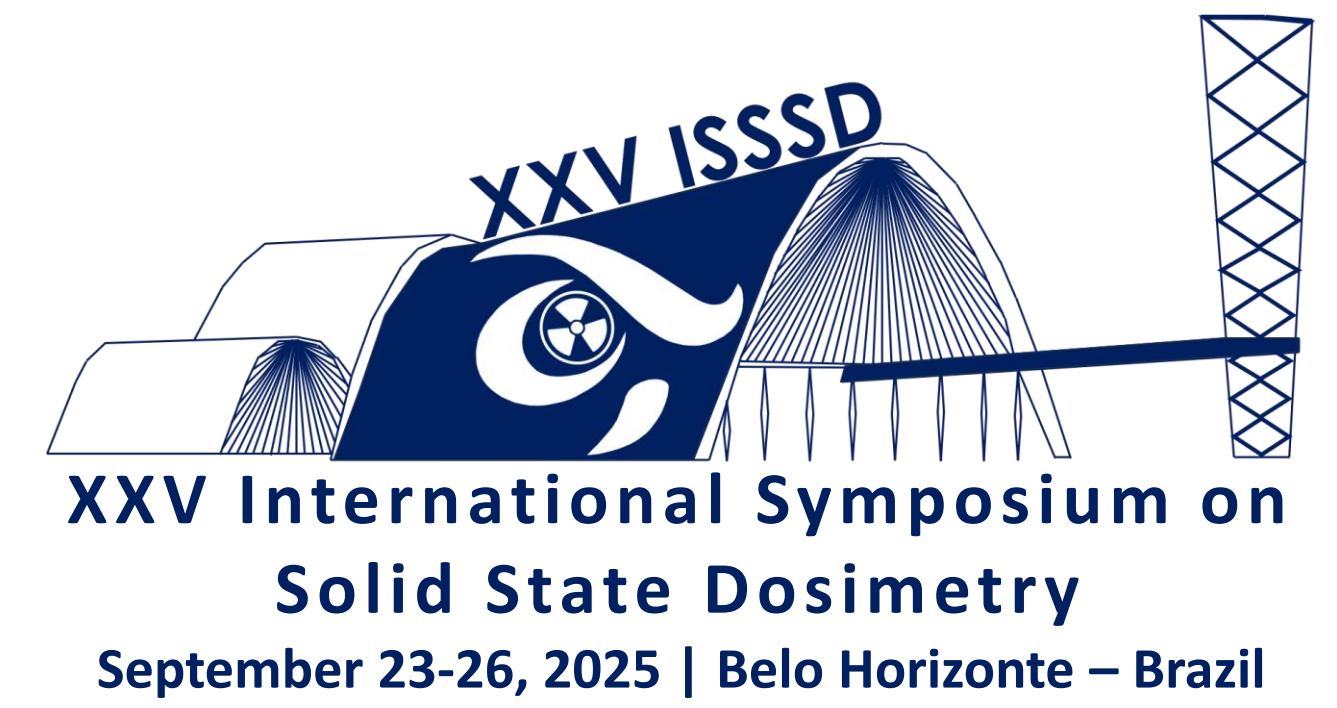


Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated $^6\text{LiI}(\text{Eu})$ Detector at the Neutron Metrology Laboratory

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1. INSTRUCTIONS

Accurate measurement of thermal neutron flux is essential for detector calibration, nuclear data validation, and radiation protection practices.

The $^6\text{LiI}(\text{Eu})$ detector is sensitive to thermal neutrons but may be influenced by epithermal and fast neutron components. Cadmium (^{113}Cd) was used as encapsulation to act as a selective filter, blocking neutrons below 0.5 eV and enabling the evaluation of the cadmium ratio (Cd-ratio). The study was carried out at the TF2 Thermal Neutron Irradiator of LNMRI/IRD, equipped with four isotopic $^{241}\text{Am-Be}$ sources (total activity of 0.6 TBq), which generate a national reference field with an approximate flux of $2 \times 10^4 \text{ n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}$ and a Cd-ratio > 200 .

2. MATERIALS AND METHODS

The experiments were carried out at the TF2 Thermal Neutron Irradiator (LNMRI/IRD), consisting of a graphite moderator ($120 \times 120 \times 120 \text{ cm}^3$) and four isotopic $^{241}\text{Am-Be}$ sources with a total activity of 0.6 TBq.

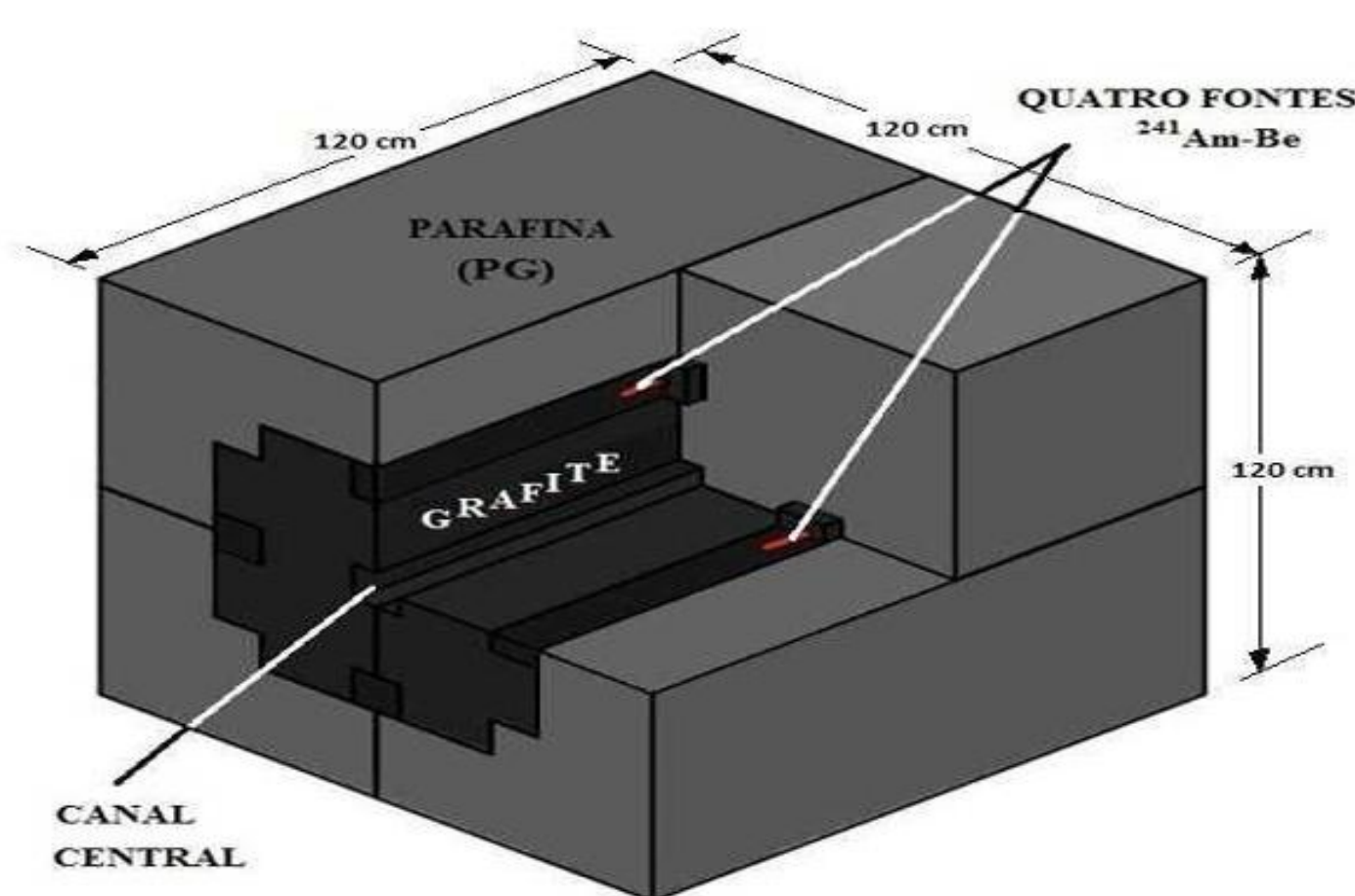


Figure 1: Thermal neutron irradiator at LNMRI/IRD, showing the graphite moderator, paraffin shielding, and arrangement of four $^{241}\text{Am-Be}$ neutron sources (adapted from Astuto et al., 2018).

The detection system employed a $^6\text{LiI}(\text{Eu})$ crystal coupled to a photomultiplier tube (PMT) and a multichannel analyzer (MCA) for pulse-height analysis.



Figure 2: $^6\text{LiI}(\text{Eu})$ scintillation detector coupled to PMT and mechanical support (LNMRI/IRD, present work).

To improve spectral selectivity, the detector was encapsulated in a cadmium (^{113}Cd) capsule with 1 mm wall thickness, 14.2 mm internal diameter, and 45.5 mm length.

Measurements were performed along the irradiation channel axis, from -10 cm to 250 cm relative to the moderator face, under both Cd-out (without filter) and Cd-in (with filter) conditions.

The cadmium ratio (Cd ratio) was determined according to:

$$\text{Cd Ratio} = \text{Cd-out} / \text{Cd-in}$$

Uncertainty analysis followed the ISO GUM methodology ($k = 2$), taking into account counting statistics, positioning accuracy, and background contributions.

3. RESULTS

Count rate profiles (Cd-in / Cd-out):

Cd-out (thermal + epithermal): showed a maximum at 30 cm, followed by a gradual decrease at larger distances.

Cd-in (epithermal + fast): remained close to background levels $\rightarrow$ confirming the high efficiency of the cadmium filter.

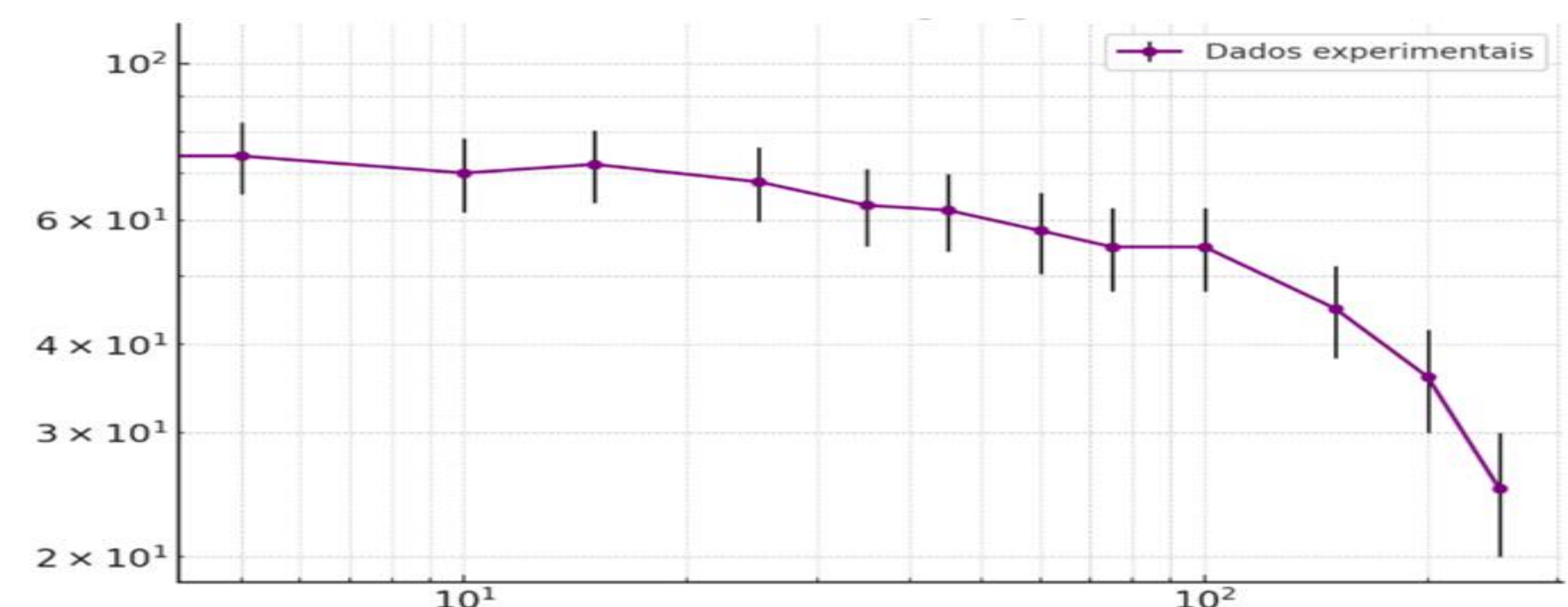


Figure 3: Normalized thermal neutron flux peaks at 30 cm (reference position) and decreases with distance due to attenuation and geometrical divergence.

Cadmium Ratio:

Maximum ~ 225 at 30 cm.

Stable values (~ 200) beyond 50 cm $\rightarrow$ homogeneous field.

Normalized Thermal Flux:

Maximum at 30 cm, decreasing with distance.

Spectral Composition:

Thermal neutrons: 97–99% of total flux.

Epithermal + fast neutrons: $\sim 1\text{--}3\%$, with slight increase near the entrance and at larger distances.

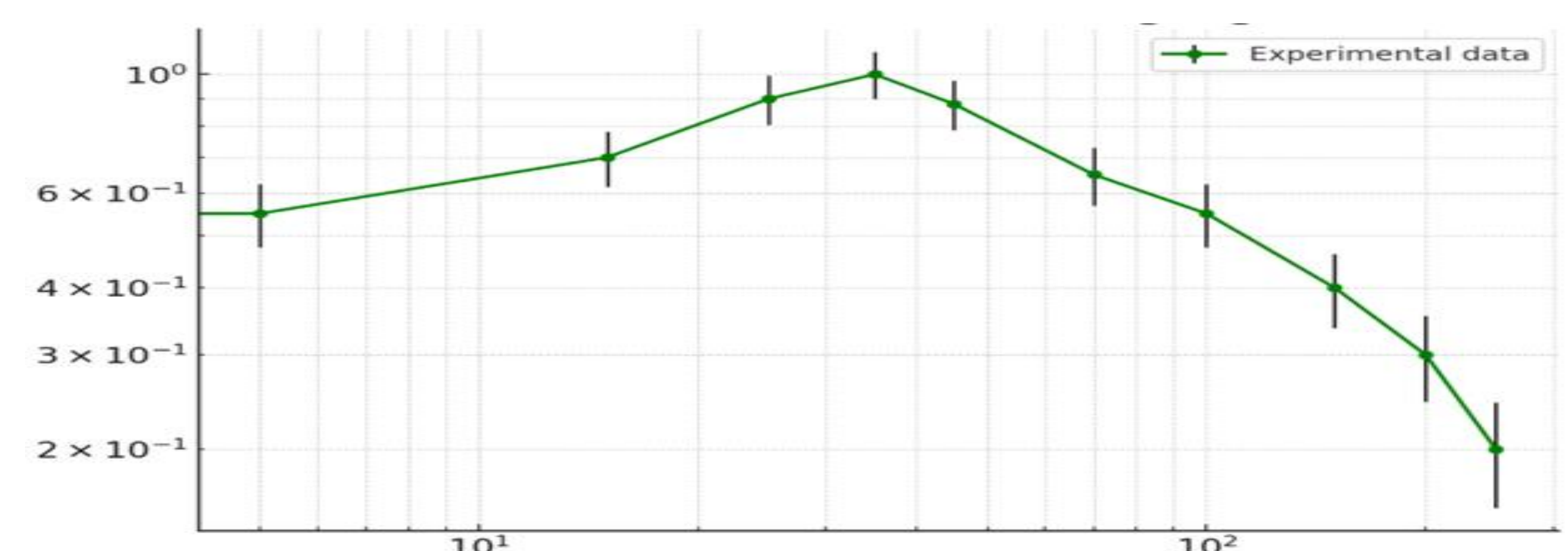


Figure 4: Thermal neutrons dominate the spectrum ($\sim 97\text{--}99\%$), while epithermal + fast components remain residual ($< 3\%$) with a slight increase at channel edges.

4. CONCLUSION

The TF2 irradiator provides a highly thermalized and homogeneous field, suitable for calibration and benchmark experiments.

Cadmium encapsulation improved selectivity and reduced interference from non-thermal neutrons.

The high Cd-ratio (> 200) confirms the predominance of the thermal component.

These results strengthen the reliability of LNMRI/IRD as a national reference in neutron metrology and contribute to applications in nuclear safety, environmental monitoring, and scientific research.

5. ACKNOWLEDGMENTS

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